

Will Langford

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ABOUT

Technical product leader with deep expertise in complex electromechanical systems, system integration, and embedded systems. Experienced taking products from early architecture through production while driving cross-functional technical decisions, product performance, and resolution of the highest-risk engineering challenges. Former startup co-founder and Head of Engineering with experience building multidisciplinary teams and bringing novel hardware platforms from prototype to deployment.

EXPERIENCE

Formlabs, Systems Integration & Embedded Systems Lead – [Fuse X1](#) 2023 - Present

- Product-level technical ownership for Formlabs next-generation industrial SLS 3D printer, spanning embedded architecture, system integration, print performance, and cross-functional technical risk from concept through mass production.
- Own embedded system architecture and integration for FuseX1, driving technical decisions and resolution of cross-subsystem issues across electrical, firmware, mechanical, thermal, sensing, controls, and print-process domains.
- Set program technical priorities through a 12+ person daily print-success triage and 18+ person weekly hardware-risk review, driving closure of 200+ print issues and the highest-risk problems blocking product ship.
- Led [Print Intelligence](#) from initial concept to production release in 6 months, developing the feasibility study and business case to secure resources, then driving data curation, algorithm and system architecture, validation, firmware integration, and release of the failure-prevention system.
- Led resolution of a program-critical print-quality issue that blocked reliable part production, developing new diagnostic methods and driving cross-functional debugging to unlock consistently good parts and enable subsequent performance optimization.

Volta Labs Inc., Co-founder & Head of Engineering 2019 - 2023

- Co-founded Volta Labs, raising \$25M in Seed and Series-A to bring to market novel electrowetting based benchtop tools to automate a wide range of workflows for DNA/RNA sequencing sample preparation.
- Designed and integrated the first prototypes capable of performing biological applications, leading the development of multiple critical IP and technology advancements in materials science, mechanical, electrical, firmware, and software domains that enable unique biological operations on the platform.
- Oversaw technical architecture development across mechanical, electrical, and software stacks, driving key product decisions that accelerated testing and iteration. Led the development of two major prototype generations, both of which resulted in multiple successful placements at collaborator sites.
- Played a pivotal role in scaling the team from 5 to 30 people, adapting the team's organizational structure and operating principles to suit the needs at each company stage along the way.
- Augmented the team's capacity and know-how through external engineering partnerships spanning materials science, electrical engineering and firmware, and mechanical design and manufacturing.
- Conducted voice of customer interviews, worked collaboratively with a cross-functional product team to refine user needs, and led a product summit to translate user needs into major product decisions.

- Fostered a culture of talent development, converting several interns and co-ops into high-performing full-time employees, through empathetic leadership and a highly collaborative environment centered on core values of bias towards action, continuous learning, and teamwork.

Makerbot Industries, *Summer R&D Associate*

Summers 2009 - 2011

- Researched and prototyped Dual-Extrusion technologies for Makerbot 3D printers.
- Designed, tested, programmed, and launched the Unicorn pen plotter tool-head.
- Supported development, production, and distribution of first generation desktop 3D printers.

EDUCATION

Massachusetts Institute of Technology, Cambridge MA

August 2012 - 2019

Doctor of Philosophy — 2019, *Discrete Robotic Construction*: [Dissertation](#) ◇ [Defense](#)

Master of Science — 2014, *Electronic Digital Materials*: [Thesis](#)

Advisor: Neil Gershenfeld — Center for Bits and Atoms

Tufts University, Medford MA

August 2008 - May 2012

Bachelor of Science in Mechanical Engineering

Overall GPA: 3.81

Minor in Engineering Management

Honors and Awards: Vincent Manno Leadership Award, The Prize Scholarship of the Class of 1882, OLeary Design Award, Mead Jonathan Taylor Prize, Summa Cum Laude, Tau Beta Pi, Deans List (all semesters)

RESEARCH PROJECTS

A Discrete Approach to Robotic Construction

2014 - 2019

My research explores assembly-based fabrication methods that enable the construction of a wide variety of robots from a small set of millimeter-scale parts.

Digital Material Assembler

2014 - 2016

I developed an automated assembly machine that is able to build electronic components by placing individual conductive and insulating building blocks. [Project link](#)

Desktop Digital Fabrication Tools

2014 - 2019

I've designed and built a number of custom desktop-scale digital fabrication tools that I use in my research.

[Desktop Wire EDM](#) ◇ [Micro-Materials Tester](#) ◇ [Punch Press](#) ◇ [Foldafab](#) ◇ [Handheld CNC](#)

Electronic Digital Materials

2012 - 2014

My masters work demonstrated and characterized the assembly of complex electronic functionality from a standardized set of conductive, insulating, resistive, and semiconducting building blocks. [Project link](#)

TECHNICAL SKILLS

Embedded systems & controls · System architecture & integration · Robotics & mechatronics · Digital fabrication / additive manufacturing · Python · Hardware/software prototyping · Sensing & instrumentation

TEACHING & LEADERSHIP

Deployed [Fablabs](#) in Saudi Arabia, [Armenia](#), [Rwanda](#), and [Bhutan](#), 2014-2018

Teaching assistant for multiple graduate courses at MIT, 2013-2018

[How to Make \(Almost\) Anything](#), [How to Make Something That Makes \(Almost\) Anything](#), [The Nature of Mathematical Modeling](#), [The Physics of Information Technology](#)

Teaching assistant for multiple undergraduate courses at Tufts University, 2011

Electronic Musical Instrument Design, Intro to Robotics and Mechatronics

Founder and president of Tufts Robotics Club, 2008-2012

Director of the Tufts Botlab, a student run robotics and fabrication lab

Led teams which placed 1st in the Trinity Firefighting Robotics Olympiad Exam twice

Granted Student Life Imagination Award for developing and conducting student robotics workshops including CNC Cupcake Frosting, Sumobot Competition, Friendly Robotics, and Toy-hacking Elmo

Mentored Medford High School and Melrose High School robotics teams

Led teams which developed an automated hydroponics gardening systems, 12lb Battlebots for Robot Conflict competitions, and a Mars rover robot

PUBLICATIONS

Langford W, Gershenfeld N. [Discretely assembled walking machines](#), Journal of Micro-Bio Robotics, 2020

Langford W, Gershenfeld N. [A Discretely Assembled Walking Motor](#), MARSS Helsinki, 2019 (**Best Student Paper Award**)

Jolly S, Savidis N, Datta B, Karydis T, Langford W, Gershenfeld N, Bove M. [Progress in fabrication of anisotropic Bragg gratings fabricated in lithium niobate via femtosecond laser micromachining](#), Advanced Fabrication Technologies for Micro/Nano Optics and Photonics XI, 2018

Langford W, Gershenfeld N. [Discretely Assembled Compliant Mechanisms](#), Proceedings of the IUTAM Symposium Architected Materials Mechanics, 2018

Langford W, Ghassaei A, Jenett B, Gershenfeld N. [Hierarchical Assembly of a Self-Replicating Spacecraft](#), IEEE Aerospace, 2017

Langford W, Ghassaei A, Gershenfeld N. [Automated Assembly of Electronic Digital Materials](#), Proceedings of the Manufacturing Science and Engineering Conference, 2016

Trinh G, Cellucci D, Langford W, Im S, Luna A, Cheung K. [Modular Rapidly Manufactured Small Satellite \(MRMSS\)](#), AIAA 2015

Patents

Hierarchical Assembly of Self-Replicating Spacecraft Using Distributed Mechanisms and Actuation in Digital Materials, [US20190077030A1](#), issued 2023

Discretely Assembled Logic Blocks [US20190190523A1](#), issued 2019

Discrete Assemblers Utilizing Conventional Motion Systems, [US10155313B2](#), issued 2018

Self-assembling assemblers built from a set of primitive blocks, [US10155314B2](#), issued 2018

Methods and systems for droplet manipulation, [WO2022187381A1](#), filed 2022

Diamond Rotors For MAS-NMR, [US20210146475](#), filed 2020

Discrete Motion System, [US20140300211A1](#), filed 2014

Electromagnetic Digital Materials, [US20140145522A1](#), filed 2013

Wireless sensor network for determining cardiovascular machine usage, [US20140300211A1](#), filed 2012

SPEAKING EVENTS

[Manipulation Automation and Robotics at Small Scales \(MARSS\) Helsinki](#), Plenary Talk, *Discrete Robotic Construction*, July 2019

[Symposium on Computational Fabrication](#), *Assembly Fabrication*, June 2018

[Dimensions of Doctor Who](#), *Using Shape-Shifting Matter To Build the TARDIS*, April 2016

[FAB11 Symposium](#), *Digital Material Assembly*, August 2015

[The Science of Digital Fabrication](#), *Micro-Assembly*, May 2013

MEDIA

[Volta Labs grabs \\$20 million to address a growing genomics bottleneck](#), Techcrunch 2022

[Volta Labs: Improving workflows for genetic applications](#), MIT News 2021 ◇ [MIT STEX Video Feature](#)

Tiny motor can walk to carry out tasks, [MIT News](#) ◇ [Digital Trends](#) ◇ [Techcrunch](#) 2019

[Adam Savage's Maker Tour](#), Tested.com 2017

[Fablabs](#), WONDROS 2016

[On GPS: Future of Digital Fabrication](#), CNN 2013

[Looking To Frost Cupcakes & Deal With the Economic Stimulus Plan? Tufts Has Some Robots For That](#), BostInno 2012

[If you build it...](#), Tufts E-News 2010

Make Magazine: [Dualstrusion](#) ◇ [Bloombot](#) ◇ [Bracelets](#) ◇ [Sumobots](#) ◇ [Glasses](#) ◇ [Braille](#)

[3-D Printers Make Manufacturing Accessible](#), Wired 2009